

Cole Uyematsu

425-445-4268 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Chess](#)

EDUCATION

Pomona College

Claremont, CA

Bachelor of Arts, Computer Science and Mathematics double major

May 2026

Thesis: Comparative Analysis of Generative Adversarial Network Architectures using Distance Metrics

EXPERIENCE

Pomona College

Claremont, CA

HCI Research Assistant

Apr 2026 – Present

- Conducting data analysis and visualization for an HCI study on shared gaze and remote collaboration under Prof. Alexandra Papoutsaki, examining how participants' communication channel affects collaborative search across 18 group sessions, 13,824 trials, and 25M+ gaze samples.
- Built an end-to-end Python (NumPy, pandas, matplotlib) data pipeline covering ingestion, eye-tracking extraction, trial alignment, and figure generation, producing 218 publication-ready figures.

Computer Science Teaching Assistant

Jan 2026 – May 2026

- Facilitated mentor sessions and graded coursework for Advanced Algorithms, covering network flow, approximation algorithms, and algorithms for incomplete information settings.

ASPC | [Website](#) | [Repository](#)

Claremont, CA

Software Developer

Aug 2025 – May 2026

- Developed a ranked-choice voting platform, achieving 47% voter turnout (786 of 1,672 students).
- Architected a bulk data ingestion pipeline integrating an external course API with MongoDB, automating semesterly updates for 1,500+ course listings.
- Redesigned course review submission flow in Next.js with improved search, growing total reviews by 2.7×.

Software Developer

Dec 2024 – May 2025

- Migrated a legacy Ruby on Rails application to MERN stack (MongoDB, Express, React, Node), improving site performance and scaling to a peak of 21,000+ monthly page views.
- Coordinated development using Jira for sprints, GitHub Actions for CI/CD, and Jest for testing, with enforced code review.

P-ai

Claremont, CA

Project Lead

Sept 2024 – May 2025

- Built backend infrastructure for *p-RoomMatch*: designed the database schema, implemented WebSocket-based real-time chat, and developed an ML clustering pipeline connecting model outputs to the frontend.

PROJECTS

Hoops Data | *NBA Analytics Platform* | [Website](#) | [Repository](#)

May 2025 – Present

- Built and shipped a full-stack NBA reference and analytics platform (Next.js, PostgreSQL) serving 500+ monthly users, with player comparison tools, historical trend analysis, and daily data-driven trivia games.
- Compiled and enriched a database of 1.6M+ box scores spanning 75+ years of NBA history using pandas and Beautiful Soup, building the ingestion pipeline that powers all site statistics.
- Engineered a daily 'Six Degrees' game linking two players through shared teammates by modeling the league as an adjacency list and running BFS to find the shortest connection path.
- Developing an agentic workflow that combines the dataset with trending storylines to draft, refine, and publish weekly NBA articles under human review.

PastPic | *Location-Based Photo Memory App (Beta)*

May 2026 – Present

- Built a cross-platform mobile app (React Native, Expo, TypeScript) that geofences locations where a user took photos and surfaces those memories when they return; shipped to TestFlight with 265 unit tests across 14K+ lines.
- Implemented a background geofencing task that runs even when the app is killed, rotating monitored regions to work around the OS 20-region limit, with adaptive trigger distances and notification filtering.

SKILLS

Languages: TypeScript/JavaScript, Python, SQL, Java, Go, HTML, CSS

Frameworks & Tools: React, Next.js, Express, FastAPI, PostgreSQL, MongoDB, Docker, AWS, Git, GitHub Actions, Cursor, Claude Code

Data & ML: Pandas, NumPy, Beautiful Soup, Scikit-Learn, PyTorch, Matplotlib

Methods: REST API Design, Data Ingestion Pipelines, CI/CD, Jest, Agile/Scrum